

Motor Control
Competence
Center

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INDUSTRIAL SUMMIT 2022
SHENZHEN, CHINA | 3 NOVEMBER
POWERING YOUR SUSTAINABLE INNOVATION

Multiaxis servo drives Innovative architectures & solutions

Seven ZENG

Motor Control Competence Center

STMicroelectronics

Motor Control
Competence
Center

Motor Control Competence Center major focus sectors

Home appliances & air con



High energy efficiency - high power
New algorithms - AI

Motor drives & servo drives



Position control
Functional safety - predictive maintenance

Power tools & high-end consumer



High speed motors
High integration

STSPIN · STPOWER · STM32 · STDRIVE · VIPerPlus · ST Op Amps · ST PROTECT

Market size

Major home appliances: 2022 TAM 4.2B$*

Motor drive: 2022 TAM: 5.1B$*

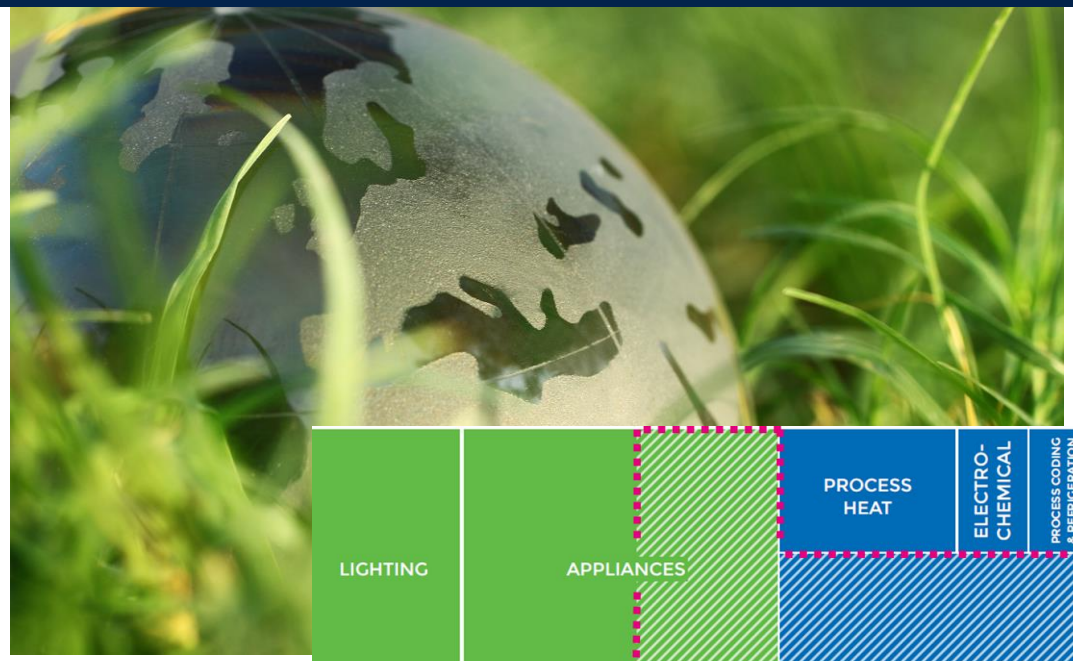
Power tools: 2022 TAM 1.2B$*

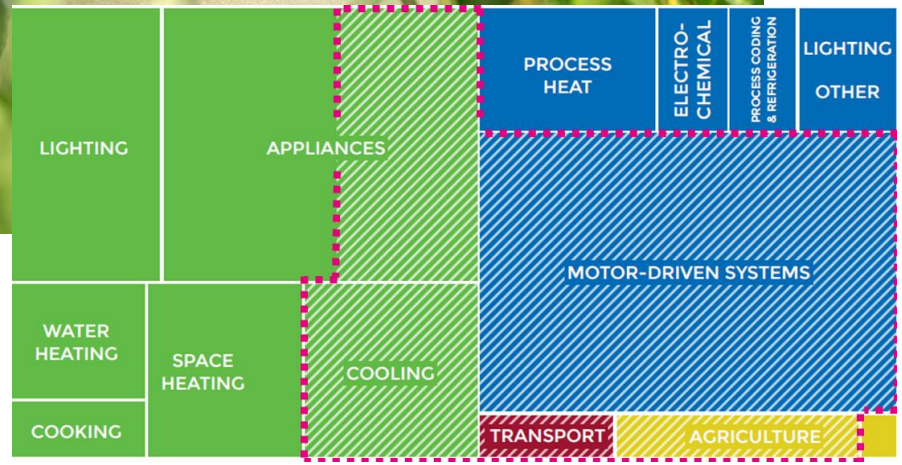
* source: OMDIA: Industrial Semiconductor Market Tracker Q32022

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Motor Drives
Energy saving jump from IE2 to IE3

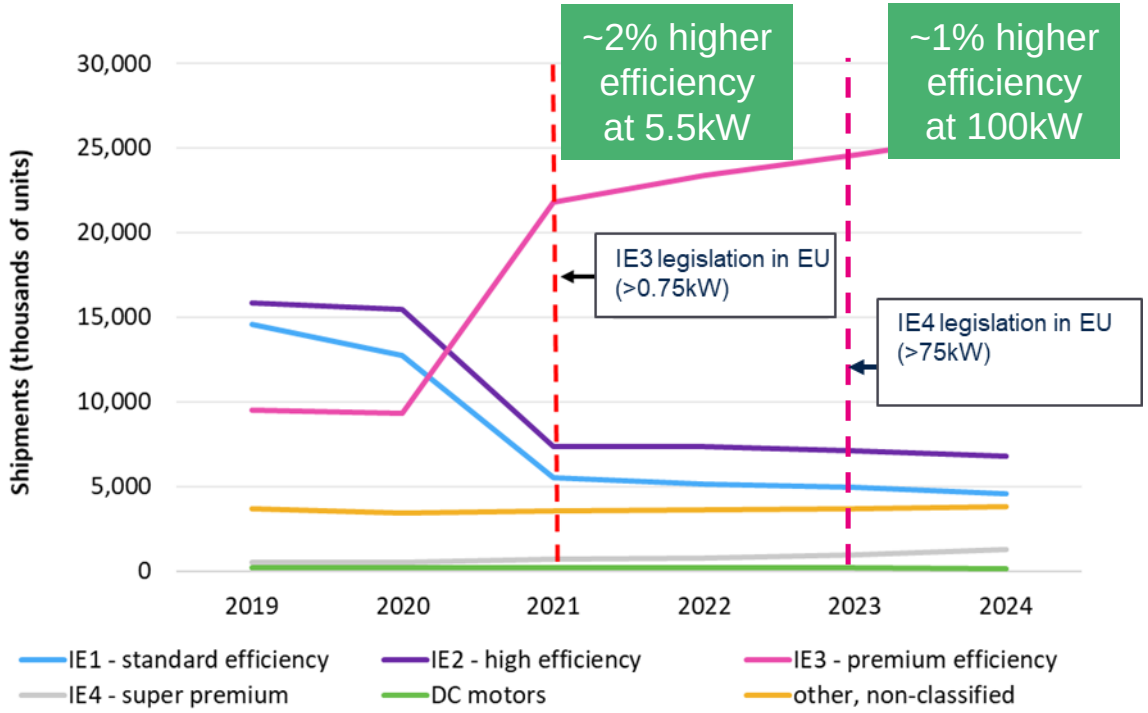
Motors account for 53%* global electricity consumption and offer the highest potential contribution to sustainability





*source: IEA, WEO 2017

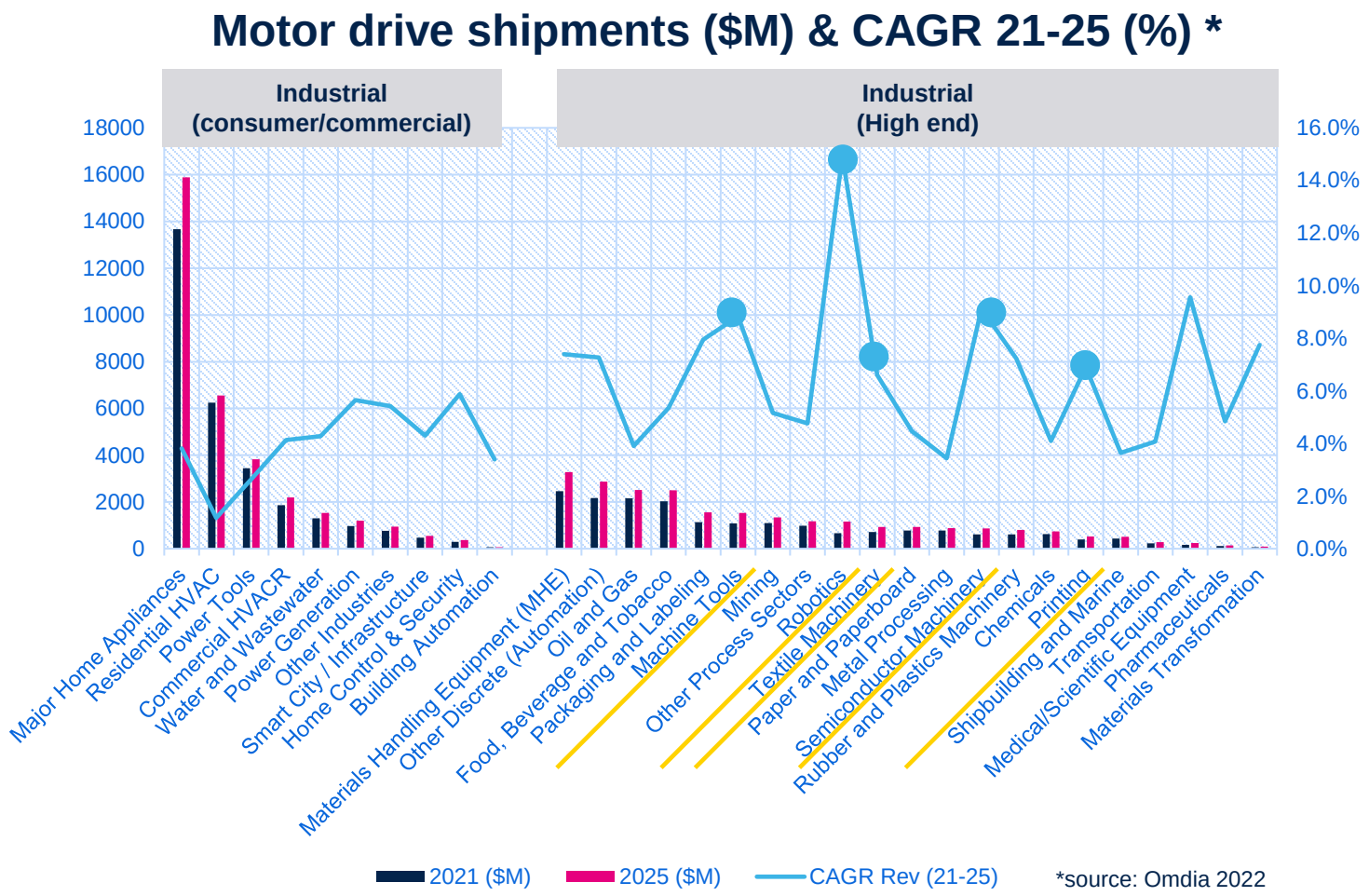
Motor drive shipment evolution by type **



**source: Omdia 2021

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Multiaxis servo drives top CAGR in high-end industrial



Robotics
CAGR 14.9%



Semiconductor machinery
CAGR 9.1%

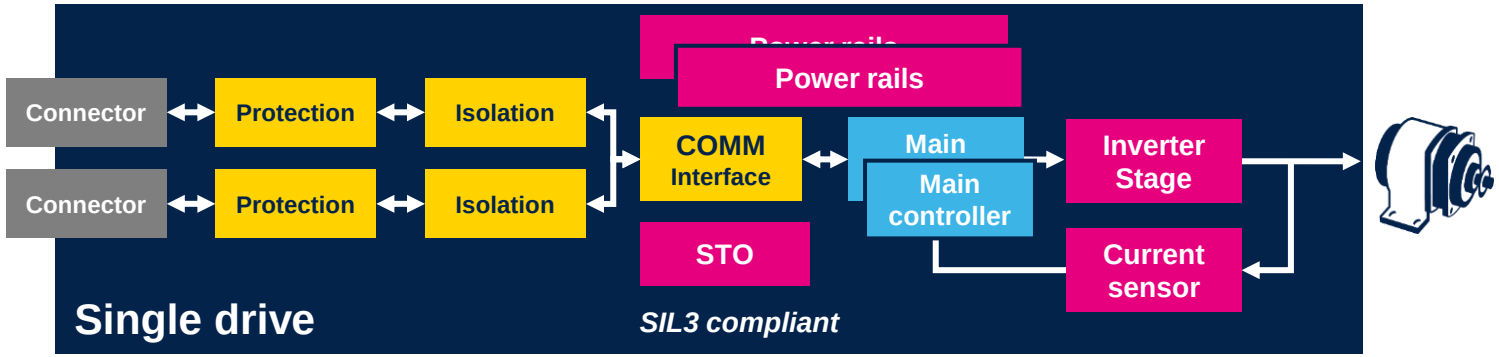


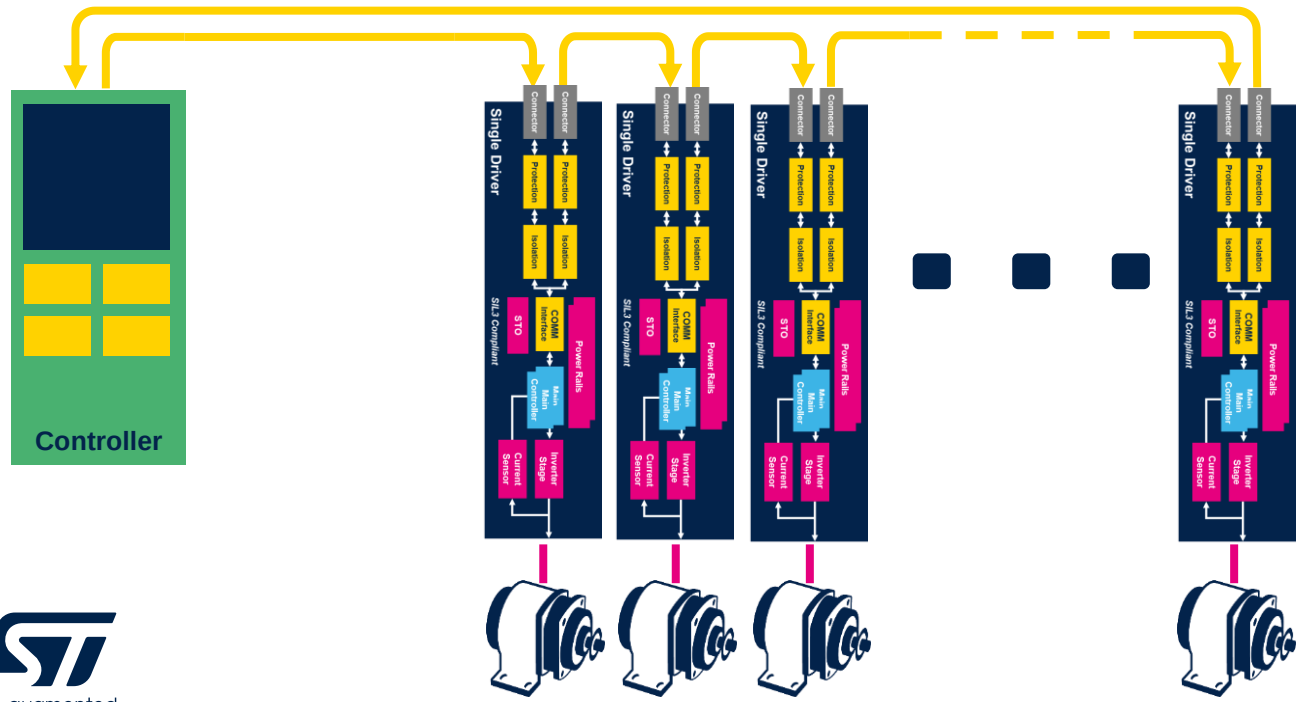
Machine tools
CAGR 8.8%



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PMSM servo drive
State-of-the-art architectures, SIL2, SIL3





Main controller
- General MCU, DSP, or FPGA

Power management
- AC-DC, DC-DC, LDO
- Inverter stage: gate drivers, power devices
- Current sensor: shunts, ICS, sigma-delta modulators
- Safe torque off

Connectivity
- Connector
- Protection from transient voltage, surge, ESD
- Isolation for human safety
- Comm interface: EtherCAT, Ethernet/IP, CAN, RS485

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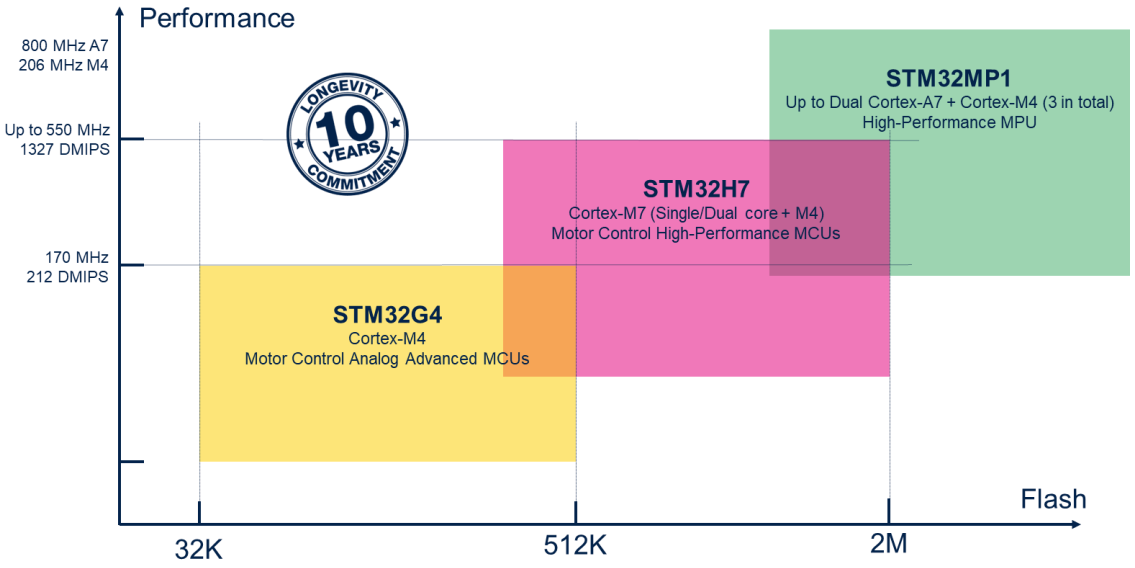
How can you innovate a multiaxis servo drive system?



New ST dual motor solution

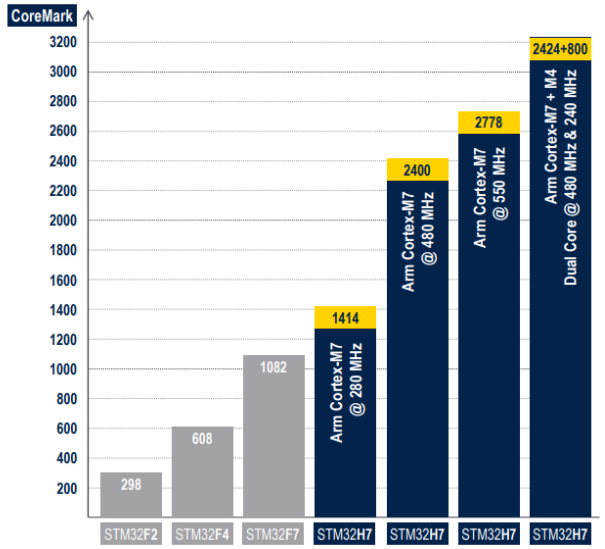
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STM32 for motor control



23% CPU load @16KHz dual FOC on STM32G431

STM32H7



Arm® Cortex® -M7 up to 550 MHz

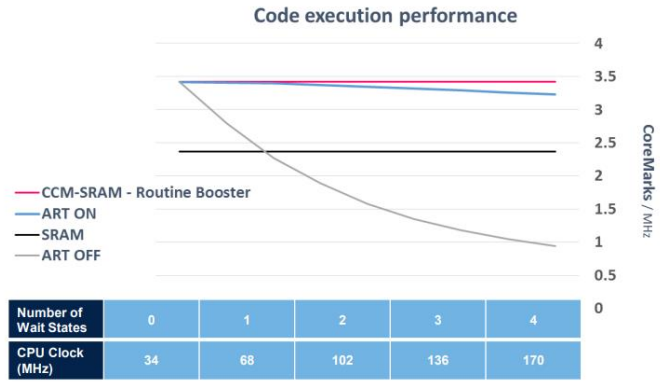
Most powerful Cortex core with double precision FPU, MPU, advanced DSP and L1 cache

Arm® Cortex® -M4 @240 MHz

Best in class core for **real-time** with single precision FPU, DSP, MPU and ART Accelerator™

STM32G4

Pure 170 MHz CPU performance (Arm® Cortex®-M4) with three accelerators



Arm Cortex-M4 with **FPU**

Up to 170 MHz CPU frequency

Up to 213 DMIPS and 550 CoreMark® results

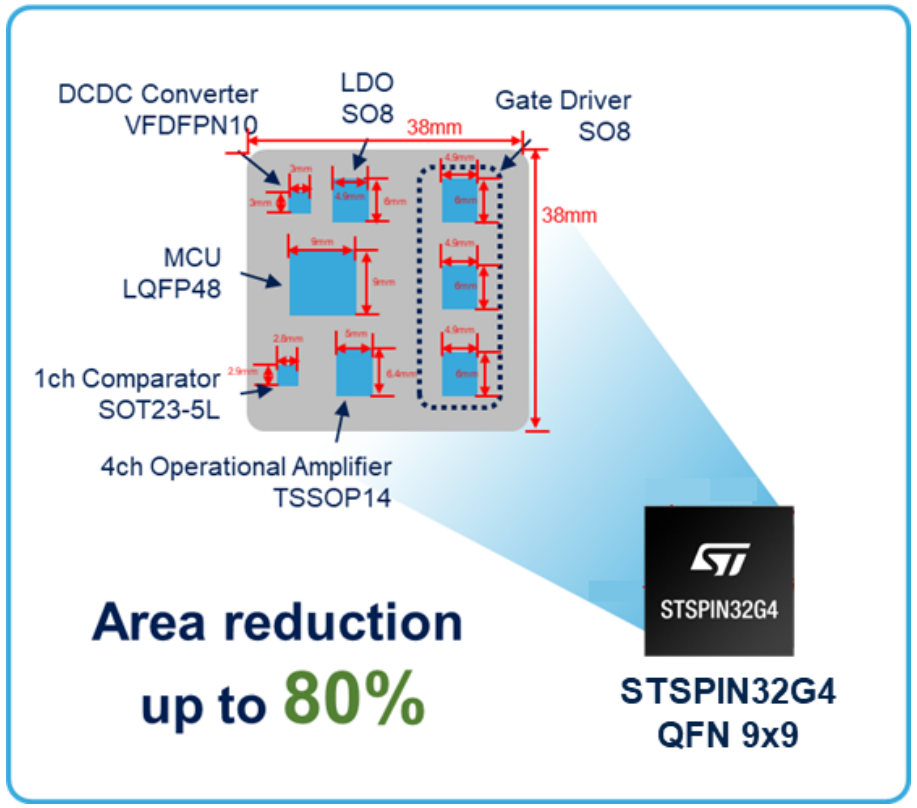
3 different **HW accelerators**:

- **ART accelerator** (~dynamic cache) → Full code acceleration (average)
- **Routine Booster** CCM-SRAM (~static cache) → determinism preserved
- **Mathematical** (Cordic + FMAC) NEW

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STSPIN32G4
5 ICs in 1 high integration motor control SiP

STSPIN32G4 vs Discrete Solution



High integration

STM32G4 + gate drivers +
Power management
~80% space reduction



Ultralow power BOR

Down to 15µA
lowest consumption

High performance 3-phase gate driver

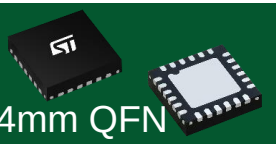
5.5V-75V
1A driving capability

QFN 9x9 mm
Up to 40 GPIO

Smart driver

VDS monitoring of the power stage
Programmable VCC output
Protections: SC,UVLO,OL,thermal

Dual motor FOC ready
+ STDRIVE101 6-in-1 gate driver



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US patents on multimotor drive current sensing network optimizations

(12) **United States Patent**
Costanzo et al.

(10) **Patent No.: US 11,105,836 B2**
(45) **Date of Patent: Aug. 31, 2021**

(54) **INVERTER AND METHOD FOR MEASURING PHASE CURRENTS IN AN ELECTRIC MACHINE**

(71) Applicants: **STMicroelectronics S.r.l.**, Agrate Brianza (IT); **STMicroelectronics (Shenzhen) R&D Co. Ltd**, Shenzhen (CN)

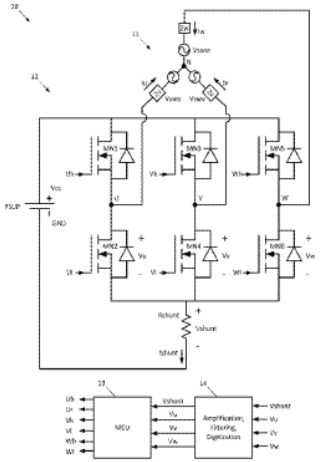
(72) Inventors: **Dino Costanzo**, Catania (IT); **Cheng Pan Cai**, Shenzhen (CN); **Xi Yu Xu**, Shenzhen (CN)

(52) **U.S. Cl.**
CPC ***G01R 19/25*** (2013.01); ***G01R 15/146*** (2013.01); ***H02M 1/00*** (2013.01); ***H02M 7/53871*** (2013.01); ***H02P 27/08*** (2013.01); *H02M 2001/0009* (2013.01)

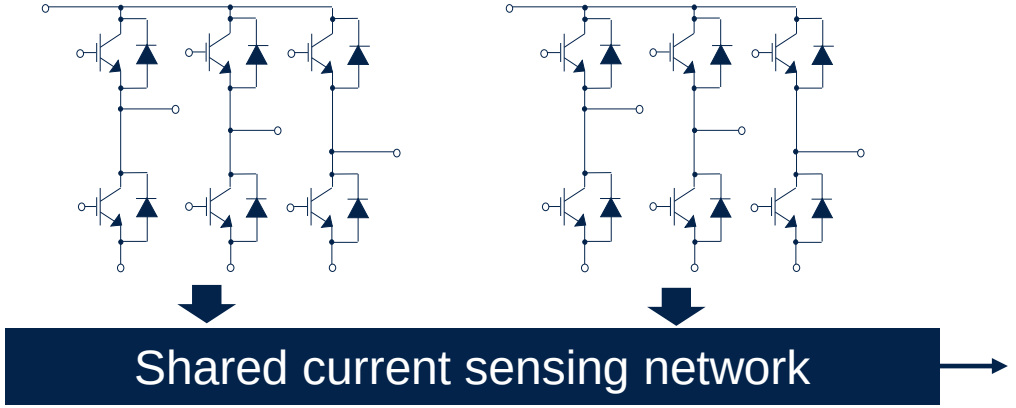
(58) **Field of Classification Search**
None
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS



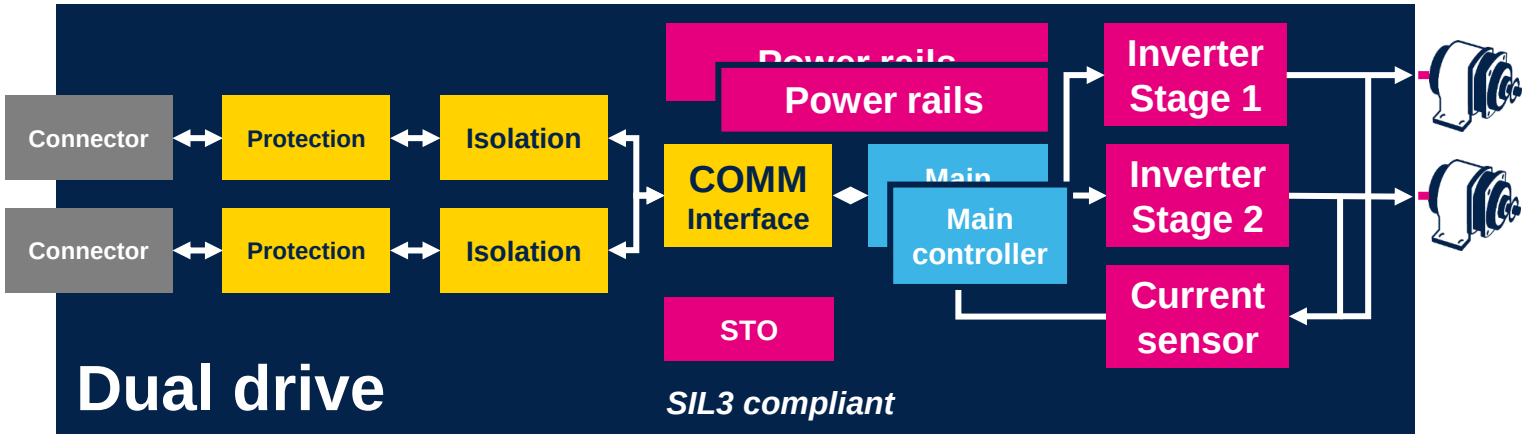
New ST single shunt current sensing breakthrough:
- **No PWM distortion**
- **Simultaneous sampling of two phase currents**

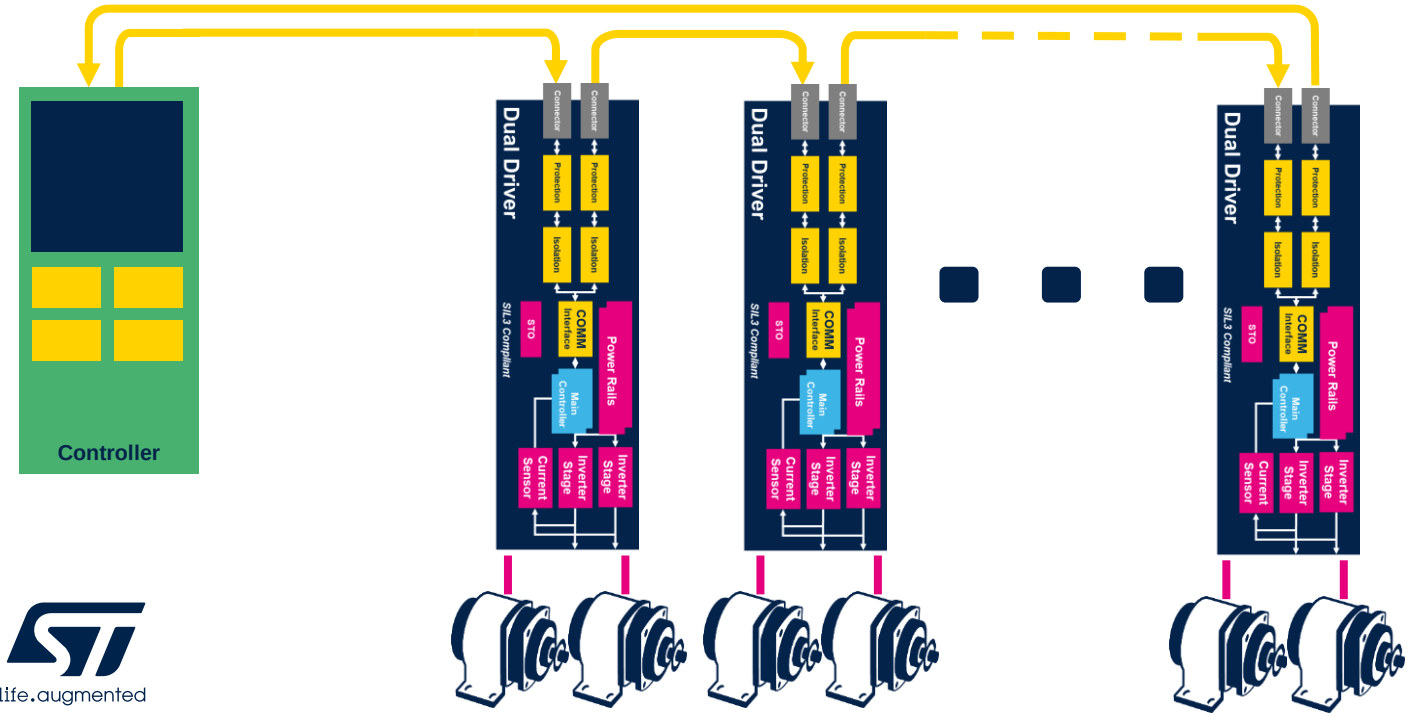


Patent pending ST multimotor shared current sensing with shunt resistors/ICs
- **Reduces current sensors & network**
- **Reduces PCB space**
- **Reduces MCU pinout assignment**

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Innovative SIL2/3 dual motor servo drive solution





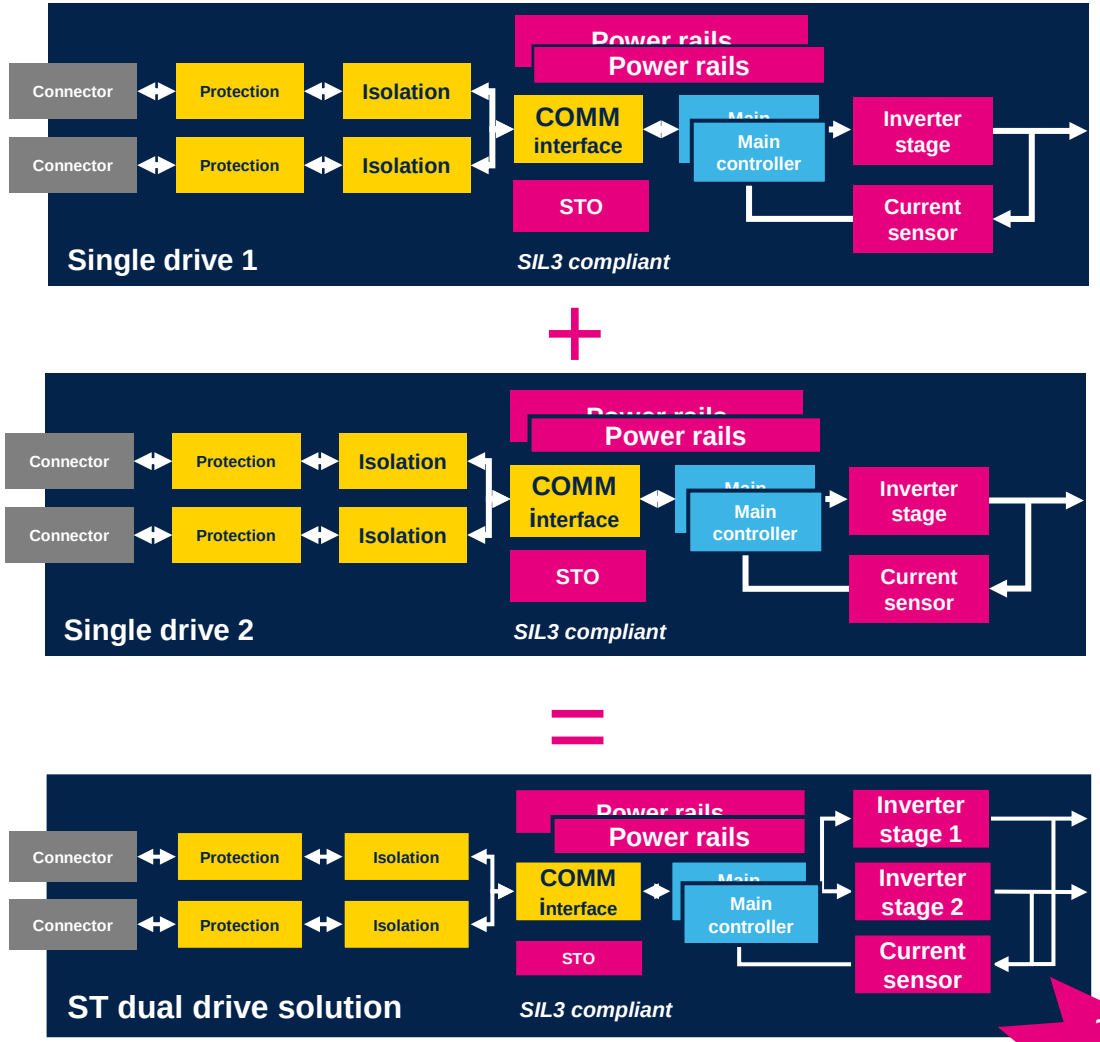
Main controller
- STM32G4; STM32H7; STM32MP1
- STSPIN32G4; X-CUBE-STL

Power management
- Power rails:
  - VIPer series, K5 series MOSFET, L7983, LDLN025M, IPS161H
- Inverter stage:
  - ACEPACK PIM, SLLIMM IPM, M Series IGBT, STDRIVE gate driver, STGAP galvanic isolated gate driver,
- Current sensor
  - ISOSD61 isolated sigma-delta modulator, TSVxxx high bandwidth op amps

Connectivity
- Protections:
  - SLVU2.8-4A1/8A1, ESDCAN04-2BWY, USBLC6-2SC6/4SC6/HSP061-4M10, ESDA14V2SC5/6,
- Comm interface:
  - ST25x NFC, BlueNRGx BLE

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Comparing architectures



~40% cost saving

Example: SIL 3 architecture / 1.5 kW / 310 VDC 220V

Component	Price (average price for 10KU)	Standard architecture servo motor drive x 2 (pcs / total price)	New ST solution dual servo drive (pcs / total price)
RJ-45	$3	4 / $12	2 / $6
Protection device	$1	4 / $4	2 / $2
Isolation	$1	4 / $4	2 / $2
EtherCAT Slave controller	$7	2 / $14	1 / $7
MCU	$10	4 / $ 40	2 / $ 20
IPM + gate drivers	$10	2 / $20	2 / $20
Power rails	$6	4 / $24	2 / $12
Shunts / ICS	$1	6 / $6	3 / $3
Total~		**$124**	**$72**

Lower cost

Higher reliability

More compact

Higher performance

SDK ready

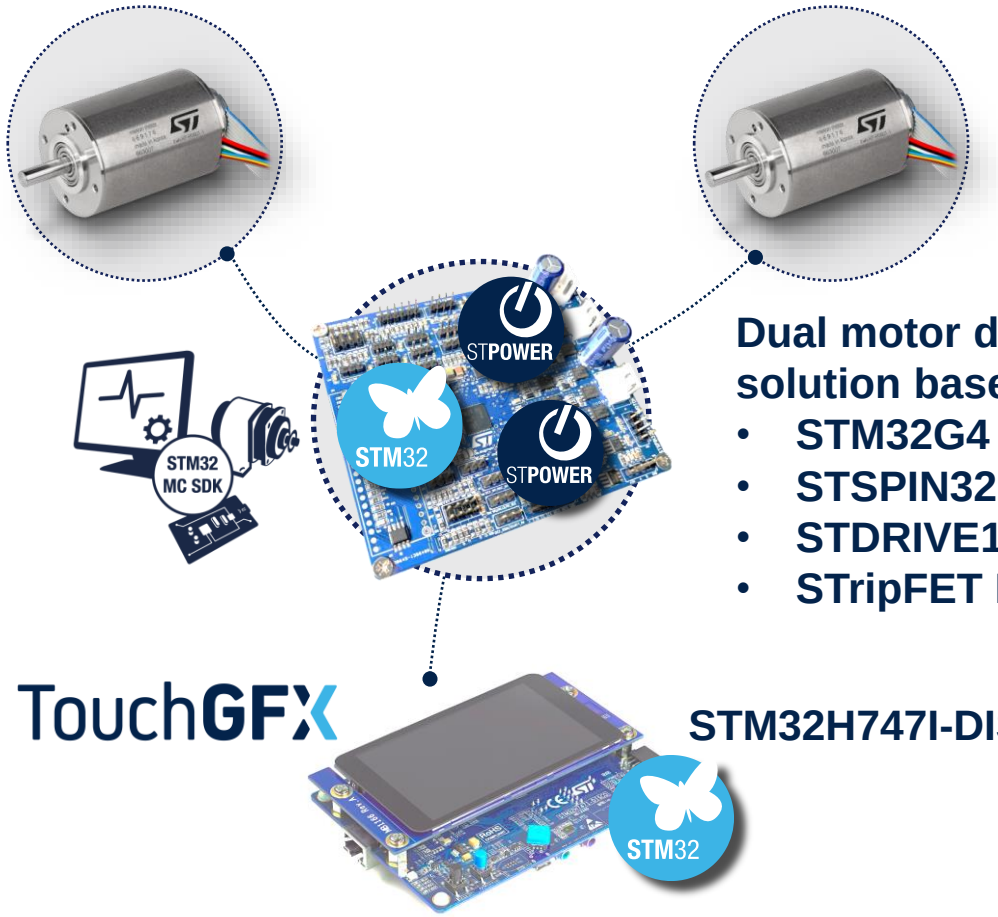
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Dual motor precision position control demo

PMSM servo

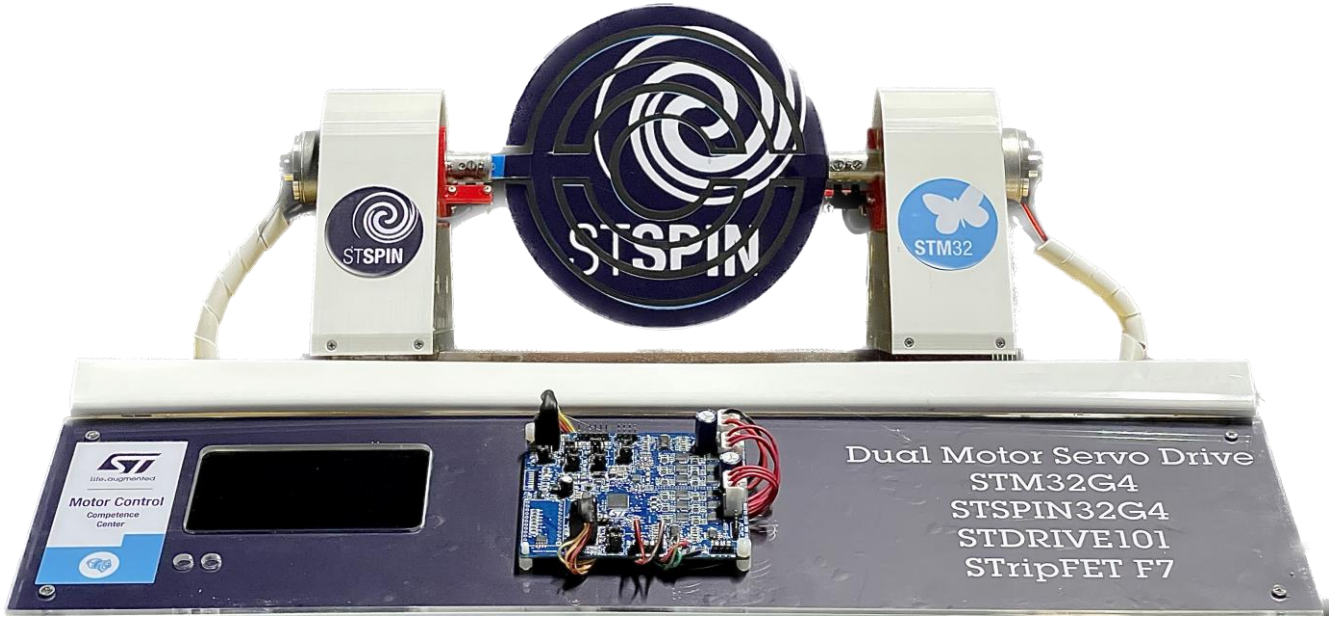
PMSM servo



Dual motor drive solution based on

- **STM32G4**
- **STSPIN32G4**
- **STDRIVE101**
- **STripFET F7**

STM32H747I-DISCO



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Our technology starts with You



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